

CUSTOM SKILLS / FIVE LIVE DEMOS

From Tribal Knowledge to Code

**Custom Skills for
GitHub Copilot Modernization**

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One app. One skill. Five demos

Create a skill. Apply it. Test its limits. Reuse it



DEMO 1 / TRY THE TASK

Modernize the library, keep the behavior

BASELINE CAPTURE: NOT-RUN

Newtonsoft.Json

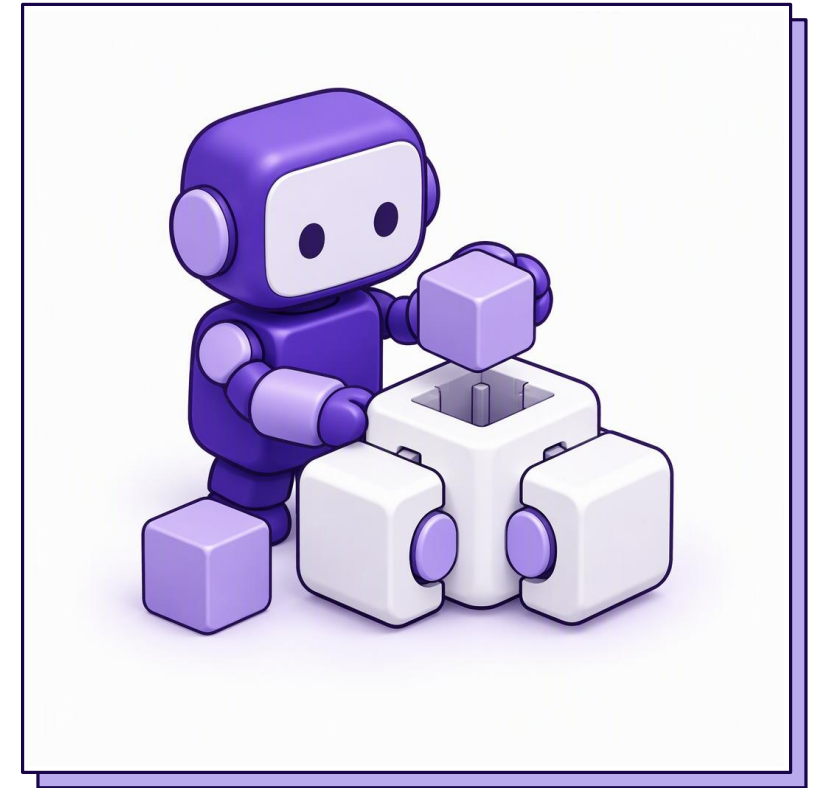


System.Text.Json

Working input

Inspect actual changes

A correct baseline is a valid result



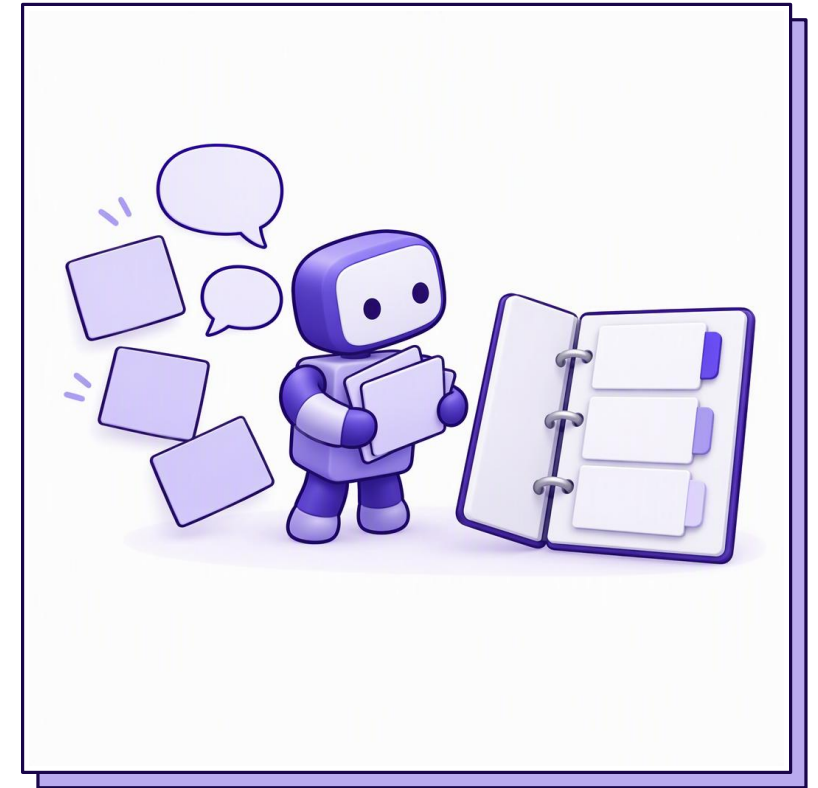
Start with the explanations your team repeats

R1 Pricing exceptions

R2 JSON compatibility

R3 Approved audit path

R4 Safe diagnostics



Real requirements need examples and observable checks

Build a reusable package

SKILL.md

name: northwind-modernization

When to apply it

Workflow and references

What evidence to report



Keep the workflow focused and its references portable

DEMO 2 / REVIEW BEFORE REUSE

When should it apply?

What would convince us?

Description	Relevant task and scope
Requirements	Concrete rules and examples
Workflow	Migration, repair or reuse
Evidence	Real checks and limitations

A description supports selection, not guaranteed adherence

DEMO 3 / APPLY THE SKILL

Same task.

Explicit team context

GUIDED CAPTURE: NOT-RUN

18 matching input files

+ 4 reviewed skill files

Fresh conversation. Explicit request

Inspect the real diff and run the same checks

Assess



Plan



Change



Review

DEMO 4 / INTRODUCE A TEACHING BUG

One option.

A broken contract

THE CONTRACT

"review_note": null

Required field
Optional value

THE TEACHING BUG

WhenWritingNull

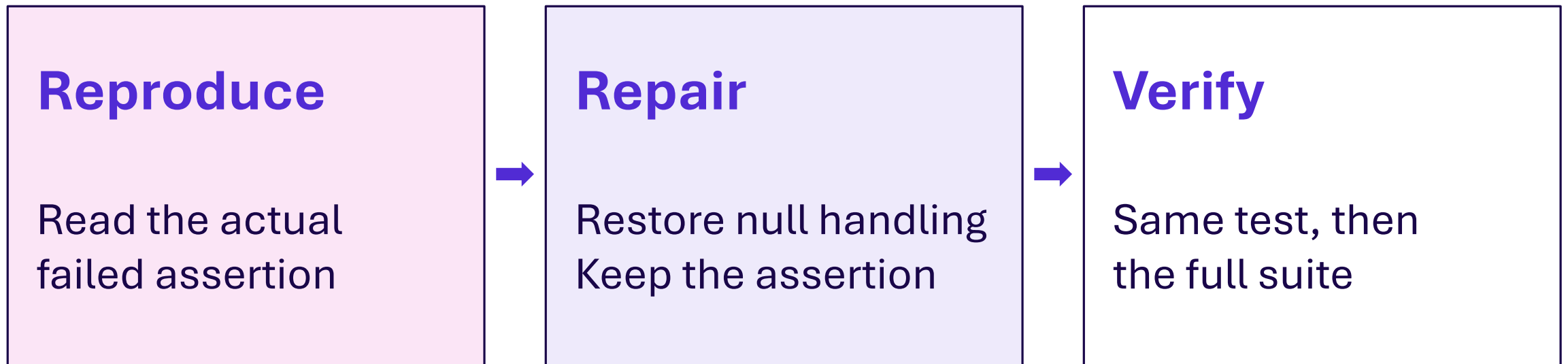
Property disappears
The assertion catches it

Deliberately introduced, not a captured Copilot mistake

DEMO 4 / REPAIR AND VERIFY

Keep the test.

Repair the implementation



Prepared red / green exercise, not agent-performance evidence

Skills carry intent. Tests detect defined regressions

DEMO 5 / REUSE THE KNOWLEDGE

A new task.

The same contract

Two existing responses



One JSON array

Same output options

Same per-item contract

Order stays intact

Empty input: []

Null input: reject

No changes to CLI, pricing or audit

Reuse output options, not another wall of instructions

RECAP / WHAT YOU JUST USED

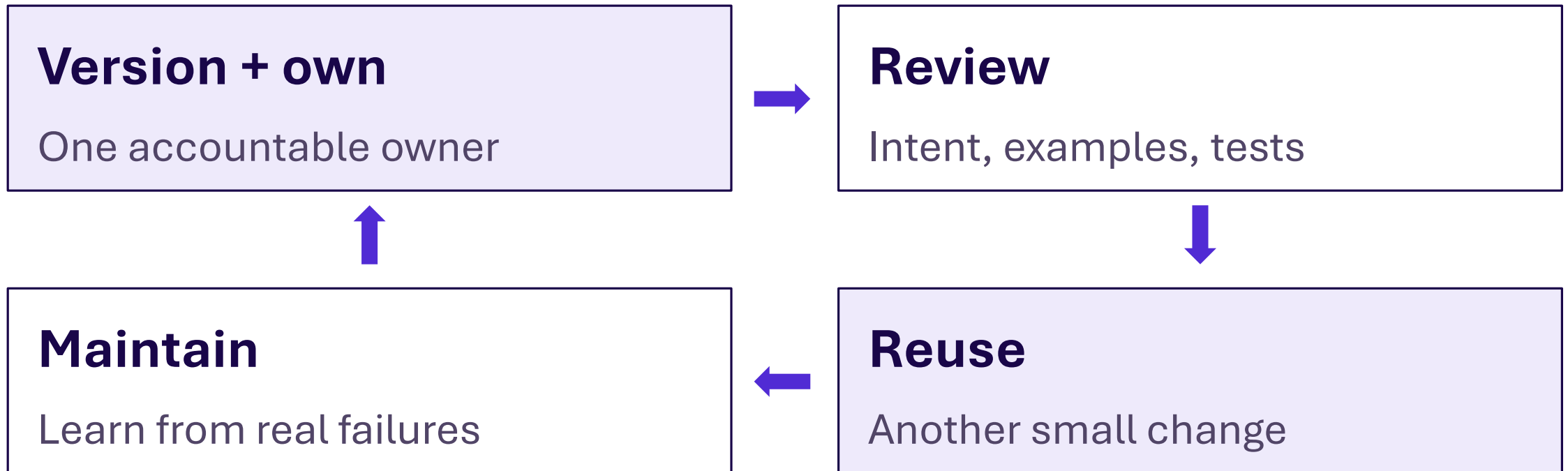
Four tools, different jobs

Prompt	Immediate task
Instructions	Broad project conventions
Skill	Relevant workflow and resources
Tests	Observable behavior

Context guides the work; evidence supports the review

YOUR NEXT STEP / ONE REPEATED EXPLANATION

Make it a team practice



Pick one task. Capture one rule. Check it. Reuse it

Try the five-demo workflow

The complete local demo kit

github.com/elbruno/GitHubCopilotModernizationSamples

Official agent skills docs

docs.github.com

Q&A

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